Practical No. 2

Title

Optimization of Multivariable Functions Using Hessian Matrix

Aim

To find and classify the stationary points of a multivariable function using partial derivatives and the Hessian Matrix.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- SymPy Library

Theory

A multivariable function is a function involving two or more independent variables. Optimization of such functions involves finding the points at which the function attains a local maximum, local minimum, or saddle point.

For a function:

$$f(x, y)$$

the stationary points are obtained by solving:

$$\frac{\partial f}{\partial x} = 0 \frac{\partial f}{\partial y} = 0$$

The nature of the stationary point is determined using the Hessian Matrix:

$$H = \left[f_{xx} f_{xy} f_{yx} f_{yy} \right]$$

where:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} f_{yy} = \frac{\partial^2 f}{\partial y^2} f_{xy} = \frac{\partial^2 f}{\partial x \partial y}$$

The determinant of Hessian Matrix is:

$$D = f_{xx} f_{yy} - (f_{xy})^2$$

Decision Criteria

- If $D > 0$and $f_{xx} > 0 \Rightarrow$ Local Minimum
- If $D > 0$and $f_{xx} < 0 \Rightarrow$ Local Maximum
- If $D < 0 \Rightarrow$ Saddle Point
- If $D = 0 \Rightarrow$ Test Inconclusive

Algorithm

1. Define the multivariable function $f(x, y)$.
2. Compute first-order partial derivatives.
3. Equate the partial derivatives to zero.
4. Solve the equations to obtain stationary points.
5. Compute second-order partial derivatives.
6. Form the Hessian Matrix.
7. Calculate the Hessian determinant.
8. Apply Hessian conditions to classify the stationary point.
9. Display the results.

Program

```
import sympy as sp


# Define variables

x, y = sp.symbols('x y')


# Define function

f = x**2 + y**2 - 4*x - 6*y + 13
```

```
print("Function:")
print(f)

# First partial derivatives
fx = sp.diff(f, x)
fy = sp.diff(f, y)

print("\nPartial Derivative wrt x:")
print(fx)

print("\nPartial Derivative wrt y:")
print(fy)

# Stationary point
critical_point = sp.solve((fx, fy), (x, y))

print("\nStationary Point:")
print(critical_point)

# Second partial derivatives
fxx = sp.diff(fx, x)
fyy = sp.diff(fy, y)
fxy = sp.diff(fx, y)

# Hessian determinant
D = fxx*fyy - fxy**2
```

```
print("\nSecond Partial Derivatives:")
print("fxx =", fxx)
print("fyy =", fyy)
print("fxy =", fxy)

print("\nHessian Determinant D =", D)

# Classification
if D > 0 and fxx > 0:
    print("\nThe stationary point is a Local Minimum.")
elif D > 0 and fxx < 0:
    print("\nThe stationary point is a Local Maximum.")
elif D < 0:
    print("\nThe stationary point is a Saddle Point.")
else:
    print("\nTest is Inconclusive.")
```

Sample Output

Function:

x**2 + y**2 - 4*x - 6*y + 13

Partial Derivative wrt x:

2*x - 4

Partial Derivative wrt y:

2*y - 6

Stationary Point:

{x: 2, y: 3}

Second Partial Derivatives:

fxx = 2

fyy = 2

fxy = 0

Hessian Determinant D = 4

The stationary point is a Local Minimum.

Result

The stationary point of the given multivariable function was determined using partial derivatives and classified using the Hessian Matrix. The stationary point (3)is a **Local Minimum**.